

CoM3D-ACE: Reciprocal Evidence-Graph Tracklet Refinement for Aerial Multi-Object Tracking

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Abstract

Aerial multi-object trackers often split one identity into several local tracklets when small targets are occluded or weakly resolved. We present CoM3D-ACE as a deterministic post-tracking refinement layer. It represents each frozen local tracklet as an EvidenceToken, builds a directed candidate graph with fixed temporal, image-plane, and appearance gates, and retains only reciprocal predecessor-successor proposals. No detector box is added, removed, or interpolated. We evaluate frozen ByteTrack, OC-SORT, and Deep OC-SORT outputs on all 50 sequences of the official MMOT test release. Relative to no refinement, equal-sequence-mean IDF1 rises by 2.28, 2.29, and 2.18 points with common oracle AABBs and by 1.65, 1.62, and 1.00 points with one shared official detector cache. Total identity switches fall by 2,545, 1,678, and 585 and by 1,389, 1,059, and 324, respectively. On the full 50-sequence release, all 12 paired sequence-bootstrap intervals against no refinement and a box-preserving AFLink adapter lie above zero. The six no-refinement intervals also remain above zero on a locally pre-specified 38-sequence reporting partition. Same-input temporal-linker controls show that the fixed geometry-first ranking is not accuracy-optimal, and a common-cost diagnostic identifies ranking as a material limitation. The fixed rule also degrades on held-out M3OT streams. The contribution is therefore an auditable, observation-preserving refinement and diagnostic study, not a claim of universal assignment optimality or cross-domain transfer.

Keywords: aerial multi-object tracking, tracklet refinement, evidence graph, reciprocal association, small aerial objects

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1. Introduction

Aerial video provides wide-area coverage for traffic monitoring, inspection, emergency response, and search-and-rescue, but its targets are frequently small in native image coordinates. In the complete MMOT test release used here, 53.4% of annotated boxes have area below 32^2 pixels and 43.1% have a short side below 16 pixels. At this scale, a short occlusion, missed detection, or weak crop can interrupt a trajectory even when the target reappears only a few frames later. The resulting identity fragmentation affects association metrics and downstream trajectory analysis without necessarily changing localization.

A target is poorly resolved when its projection contains too few sensor samples to preserve class- or identity-discriminative appearance. Enlarging a few-pixel crop cannot recover evidence that the sensor did not record. Within one UAV stream, the same target may consequently disappear and reappear under a new local identity. Figure 1 connects this weak visual evidence to the temporal identity-fragmentation problem evaluated here.

Modern online trackers attack this problem within the frame-to-frame update. ByteTrack recovers low-score detections during association [1]; OC-SORT improves motion reasoning through observation-centric updates [2]; and Deep OC-SORT supplements that design with appearance cues [3]. A distinct option is to retain these tracker outputs and link compatible tracklets after tracking. Post-processing is attractive because it can be applied to archived results and compared under exactly the same detections and local trajectories. It is also risky: one false link can merge two otherwise valid identities.

This work asks a deliberately narrow question: can a fixed evidence-graph rule improve temporal identity continuity without changing the upstream detector or tracker? We instantiate each local tracklet as an EvidenceToken, form directed candidate relations between temporally ordered fragments, and retain only mutual predecessor-successor choices that pass fixed geometry and appearance gates. A component-level frame-overlap assertion validates the relabeled output. Fragments with no admissible relation remain unchanged. The evaluated MMOT rule uses no camera pose, 3D reconstruction, ground-truth identity, learned link classifier, or box interpolation.

CoM3D-ACE begins after a local tracker has produced tracklets and ends with relabeled temporal trajectories. This separation makes the contribution tracker-agnostic and permits every method to receive exactly the same boxes,

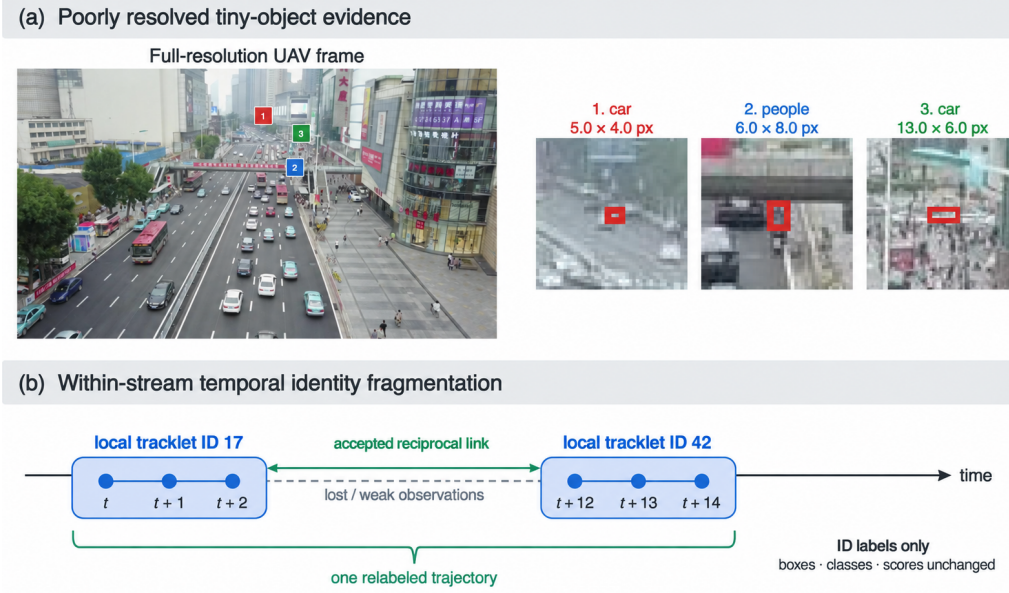


Figure 1: Tiny-object evidence and temporal identity fragmentation. (a) Representative objects in a real VisDrone frame occupy only a few native image pixels, leaving limited appearance detail even after enlargement. Box dimensions report native pixels; enlargement does not create new evidence. (b) A schematic timeline shows how one UAV stream can split a target into two local tracklets around weak or missing observations. The evaluated refiner links compatible frozen tracklets and changes only their identity labels; boxes, classes, and scores remain fixed.

classes, confidence values, and local identities before refinement.

The verified strengths of this study are:

- a tracker-agnostic deterministic refinement rule with fixed temporal, image-plane, appearance, and reciprocal constraints, together with explicit output-invariance checks and unchanged output for unsupported fragments;
- protocol-matched evaluation of three frozen tracker families under both common oracle AABBs and one shared official detector output on all 50 MMOT test sequences, with higher IDF1 than no refinement and AFLink+VA in all six conditions and all 12 paired intervals above zero; and
- a locally pre-specified 38-sequence reporting analysis whose six paired

IDF1 intervals against no refinement remain above zero, accompanied by same-input ranking, output-invariance, transfer, and runtime audits.

The contribution is an auditable post-tracking refinement layer that can be applied without detector or tracker retraining, together with a controlled account of where its fixed ranking succeeds and fails. Its positive quantitative claim is limited to MMOT improvements over unchanged upstream trajectories and over the validated AFLink adapter. Stronger alternative temporal linkers remain visible in a dedicated main comparison and paired analysis, together with transfer boundaries.

2. Related Work

2.1. *Small-object evidence in aerial perception*

Aerial detectors preserve shallow spatial detail or add high-resolution and multi-scale paths for poorly resolved targets [4, 5]. These methods improve upstream sensing. Cooperative systems can additionally combine complementary views through selective feature communication or cross-UAV feature mapping [6, 7]. CoM3D-ACE addresses a narrower stage: it starts from frozen object-level tracker outputs and asks whether temporally fragmented tracklets can be joined without changing the detector or tracker.

2.2. *Aerial tracking and evaluation*

Aerial MOT combines identity association with small target size, camera motion, viewpoint change, and crowded scenes. MDMT/MIA-Net introduced a multi-drone benchmark and an occlusion-oriented association model [8]; TSMMT models temporal and spatial interactions for multi-drone tracking [9]. MMOT provides multispectral UAV video with oriented annotations and dense, small targets [10], and M3OT adds multi-drone RGB and infrared sequences [11]. The protocol here evaluates individual temporal streams, isolating within-stream tracklet refinement from global cross-camera identity assignment.

ByteTrack associates high- and low-confidence detections [1]. OC-SORT emphasizes observation-centric motion updates under occlusion [2], and Deep OC-SORT introduces adaptive appearance association [3]. We use all three as frozen upstream trackers rather than modifying their update rules. HOTA decomposes tracking accuracy into detection and association components [12]; IDF1 evaluates identity continuity through global trajectory matching [13]. Both are reported with CLEAR identity switches.

2.3. *Post-tracking tracklet linking*

Global association can repair trajectories that an online tracker has split. GIAOTracker combines local aerial tracking with a global linking stage [14]. StrongSORT++ includes AFLink, a learned post-linker, and Gaussian-smoothed interpolation as distinct modules [15]. Our comparison reuses assignments from the released AFLink model and checkpoint, excludes interpolation, and applies a separately named output-validity adapter so that every method preserves the common box set. The native post-remap deduplication is audited separately.

Graph-based MOT methods learn structured relations among detections or tracklets. Neural message passing can optimize global associations across a detection graph [16], while city-scale multi-camera systems combine appearance and spatiotemporal constraints [17, 18]. CoM3D-ACE is narrower: it is a deterministic post-processing rule with no trainable refiner weights and a frozen pretrained descriptor; its gates are selected on declared development streams.

2.4. *Selective and constrained association*

Every post-linker trades recovered continuity against false merges. Selective prediction motivates leaving uncertain cases unresolved rather than forcing a decision [19], but a heuristic distance threshold is not automatically a calibrated risk estimate [20, 21]. We therefore report accepted-link error together with coverage at the fixed operating point and retain a descriptive threshold grid in the reproducibility artifact. The component frame-overlap check is a hard output-validity assertion; under the one-shot strict-time reciprocal graph evaluated here, it is structurally redundant rather than an independently active performance module.

3. Method

3.1. *Evaluated temporal pipeline*

Figure 2 summarizes the complete path evaluated in this paper. Frozen local tracklets are normalized as EvidenceTokens, and the CoM3D-ACE path generates temporally ordered candidates under fixed gates before relabeling identities only. All comparison linkers start from the same frozen tracklets, with their specific candidate rules defined in Section 4. Tracking metrics, paired intervals, accepted-link audits, and M3OT transfer then characterize both improvement and failure conditions.

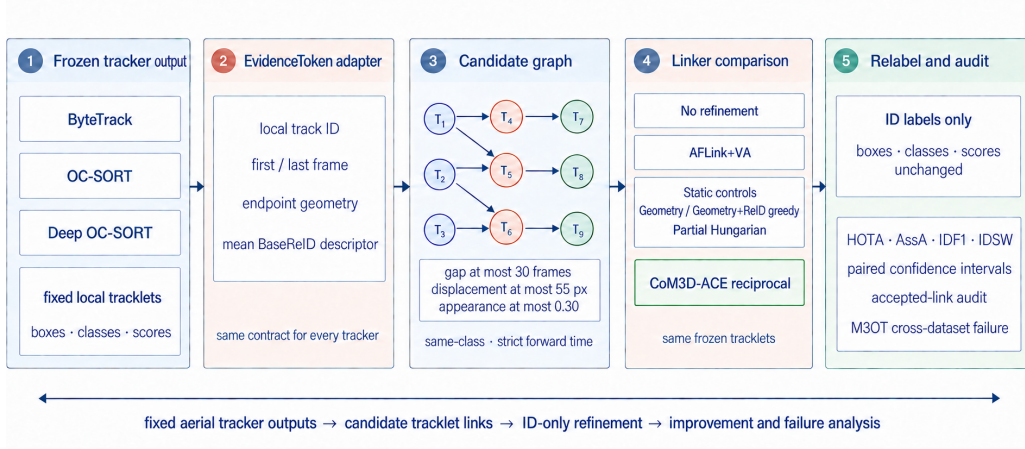


Figure 2: Evaluated CoM3D-ACE temporal-refinement pipeline. Frozen ByteTrack, OC-SORT, or Deep OC-SORT tracklets enter a common EvidenceToken adapter. Fixed temporal, image-plane, and appearance gates define the CoM3D-ACE candidate graph. Every comparison starts from the same frozen tracklets, although AFLink and geometry-only controls retain their declared candidate definitions; only the ranking/solver diagnostic shares one fixed gated graph. Refinement changes track IDs only, after which metrics and link-level diagnostics expose both improvements and transfer failures.

3.2. Frozen tracklets as evidence tokens

Let an upstream tracker produce a set of local tracklets $\mathcal{T} = \{T_i\}_{i=1}^K$. Tracklet T_i contains observations $z_{it} = (t, b_{it}, s_{it}, c_i)$, where t is the released frame index, $b_{it} = (x, y, w, h)$ is an axis-aligned box in native image pixels, s_{it} is the upstream confidence, and c_i is the class. In the evaluated adapter, classes are processed independently, so every candidate relation is same-class by construction.

An EvidenceToken for tracklet i stores T_i , its frame support $F_i = \{t : z_{it} \in T_i\}$, first and last endpoint boxes, and an optional appearance descriptor $\mathbf{f}_i \in \mathbb{R}^{128}$. The descriptor is computed from at most three direct tracker-box crops: the first, middle, and last observations. Each crop is resized to 128×256 , transformed by the fixed BaseReID evaluation normalization, and L2-normalized after encoding. We average the available crop features and normalize the result again. In the MMOT protocols, no ground-truth matching is used to select or encode a crop. The M3OT oracle-box transfer diagnostic uses a separate crop-admission procedure documented in Section 4.

The full appearance-dependent rule requires a valid descriptor at both edge endpoints. If either descriptor is unavailable because all sampled crops

are invalid, that edge is inadmissible; a tracklet with no other admissible relation retains its original identity. We do not impute a zero vector or silently fall back to geometry within the full method.

3.3. Directed candidate graph

A directed edge $i \rightarrow j$ is temporally eligible only if the two frame supports are disjoint, T_i ends before T_j begins, and

$$1 \leq g_{ij} = \min F_j - \max F_i \leq G, \quad G = 30. \quad (1)$$

Let $\mathbf{p}_i^{\text{end}}$ and $\mathbf{p}_j^{\text{start}}$ be the centers of the endpoint boxes. The image-plane and appearance distances are

$$d_{ij}^{xy} = \|\mathbf{p}_i^{\text{end}} - \mathbf{p}_j^{\text{start}}\|_2, \quad d_{ij}^{app} = 1 - \mathbf{f}_i^\top \mathbf{f}_j. \quad (2)$$

The fixed candidate gates are $d_{ij}^{xy} \leq 55$ pixels and $d_{ij}^{app} \leq 0.30$. These values were fixed from M3OT development data before the held-out tests used here. They are not adapted to sequence name, modality, tracker, or MMOT identity labels.

3.4. Reciprocal proposal and component constraint

For every source tracklet, the method chooses the outgoing candidate with the lexicographically smallest tuple

$$q_{ij} = (d_{ij}^{xy}, d_{ij}^{app}, g_{ij}, i, j). \quad (3)$$

It independently chooses the smallest incoming tuple for every destination. An edge becomes a proposal only when it is both choices. This reciprocal test prevents one source from being attached to a destination that prefers another predecessor, while naturally leaving unsupported tracklets unmatched.

Proposals are processed in ascending $(d_{ij}^{app}, g_{ij}, i, j)$ order with a disjoint-set data structure. Before joining components C_i and C_j , the method checks

$$\left(\bigcup_{u \in C_i} F_u \right) \cap \left(\bigcup_{v \in C_j} F_v \right) = \emptyset. \quad (4)$$

A proposal that violates Eq. (4) is rejected; otherwise the components are joined and inherit the earlier root identity. In the implemented one-shot graph, however, strict forward-time proposals and at most one reciprocal predecessor

and successor make every connected component a directed path. Given duplicate-free input tracklets, path members cannot overlap in time. Property tests over all 1,989 MMOT sequence/class instances therefore observe no guard rejection and no guard/no-guard output difference. Equation (4) is retained as a defensive assertion and output validator, not as an independently active performance module.

Finally, only track IDs are remapped. Frame indices, boxes, classes, confidence values, and the number of predictions are invariant. No interpolation fills temporal gaps.

This implementation does not treat the candidate score as calibrated uncertainty or issue observation commands. With E eligible edges, exhaustive candidate construction costs $O(K^2)$, proposal sorting costs $O(E \log E)$, and union-find updates are near-linear in the number of accepted proposals. The explicit candidate graph requires $O(K + E)$ memory.

4. Experimental Protocol

4.1. Datasets, inputs, and frozen split

MMOT. We use all 50 sequences in the official MMOT test release [10]: 5,466 frames and 196,194 valid annotated boxes. Raw images are HWC `uint8` arrays containing eight spectral bands. Following the MMOT paper and the active official tracking loaders, pseudo-RGB is formed from one-based bands 5, 3, and 2, corresponding to zero-based indices $[4, 2, 1]$ in RGB order. All appearance-dependent results in this paper use this $[4, 2, 1]$ mapping.

Two input conditions are kept separate. The *oracle-AABB* condition converts each official oriented annotation to its axis-aligned envelope, assigns confidence 1.0, and gives every method the same boxes and classes. Ground-truth identity remains hidden until scoring. The *official detector* condition runs the MMOT authors’ YOLO11L-3ch checkpoint once at 1,280-pixel input, confidence 0.1, and NMS IoU 0.6. Its 193,977 detections over all 5,466 frames, including one frame with no detection, are cached and reused by every tracker and refiner.

For reporting, a fixed 12-sequence partition is named `legacy12`; every other official test sequence is assigned to `confirmation38`. No sequence was selected by performance. The exact membership is listed in Table A.8. The rule and all thresholds remain unchanged across both partitions. Because no externally timestamped preregistration is available, we call this split locally pre-specified rather than independently preregistered.

M3OT transfer. M3OT [11] contributes four development streams (1-08T, 1-08, 2-08T, and 2-08) and four held-out streams (1-03T, 1-03, 2-03T, and 2-03). Suffix T denotes infrared. The temporal, geometry, appearance, and reciprocal settings were fixed from the development streams before held-out evaluation. Each stream is scored independently with the release’s vehicle filter and common oracle AABBs. To reproduce the frozen descriptor cache, tracker observations are one-to-one matched to same-frame oracle GT and admitted for descriptor extraction only when $\text{IoU} \geq 0.9$. The retained crop still uses the tracker box. Thus this transfer experiment is an oracle-box diagnostic, not a detector-input result: GT determines tracker input boxes, descriptor-crop admission, final metrics, and post-hoc link labels. We do not use the M3OT held-out result to retune the rule.

Both datasets are used for within-stream temporal identity. Neither protocol evaluates a global identity shared among different UAV cameras.

4.2. Upstream trackers and appearance model

ByteTrack, OC-SORT, and Deep OC-SORT are instantiated through BoxMOT commit `6edfa8ad` [22]. Each official fine-grained MMOT class is processed by an independent tracker instance, every released frame is passed in numeric order, and no area, height, or aspect-ratio filter is added by our adapter. Table 1 records the active settings. They are the frozen BoxMOT AABB configurations used by this study, rather than the rotated-tracker settings in MMOT’s native benchmark code.

Table 1: Frozen upstream tracker settings. A dash denotes a parameter not used by that tracker. Deep OC-SORT additionally uses its default adaptive appearance weighting and camera-motion compensation.

Tracker	Det. gate	Match gate	Max age	Min hits	IoU gate	Δt
ByteTrack	0.45	0.80	30	3	0.30	–
OC-SORT	0.30	–	30	3	0.30	3
Deep OC-SORT	0.50	–	30	3	0.30	3

Tracklet descriptors use a fixed 128-dimensional ResNet-50 checkpoint from the OpenMMLab MOT17 BaseReID configuration. Its SHA-256 is reported in Appendix E. The test transform resizes crops to 128×256 and applies the checkpoint’s ImageNet normalization. The BaseReID encoder is frozen, and the deterministic refiner has no trainable weights. The temporal, image-plane, and appearance gates were selected using the four M3OT development

streams and then applied unchanged to MMOT and the held-out M3OT streams. Neither the descriptor nor the refiner is fine-tuned on MMOT. The official detector checkpoint is MMOT-trained and is outside that statement. Deep OC-SORT receives per-detection embeddings from the same checkpoint; CoM3D-ACE then computes tracklet-level descriptors from the frozen output boxes.

4.3. Comparison methods

All methods receive exactly the same saved upstream tracklets for a given tracker and input condition. The primary operational comparison contrasts the unchanged tracker output, a released learned post-linker under the common box-preservation contract, and the proposed evidence-graph refiner:

- **None** leaves every local identity unchanged.
- **AFLink+validity adapter** reuses assignments from the official StrongSORT implementation and checkpoint [15], with released settings $\text{thrT} = (0, 30)$, $\text{thrS} = 75$, and $\text{thrP} = 0.05$. Native AFLink remaps identities and then keeps only the first row for each duplicate frame-identity pair. For the common-box comparison, a fixed no-GT adapter instead rejects an assigned component merge whenever its frame supports overlap. Gaussian-smoothed interpolation is not applied.
- **CoM3D-ACE** is the reciprocal rule in Section 3.

Three deterministic assignment controls remain a separate main diagnostic. They start from those same frozen tracklets; their candidate and ranking rules are:

- **Geometry greedy**, which takes every temporally eligible edge within 55 pixels, orders edges by endpoint distance, and applies the component frame-overlap constraint;
- **Geometry+ReID greedy**, which applies the same temporal, geometry, and appearance gates as CoM3D-ACE, then processes all gated edges by appearance distance, gap, and deterministic tracklet-ID tie breaks; and

- **Partial Hungarian**, which uses candidates satisfying the same geometry and appearance gates and a cost equal to the mean of $d^{xy}/55$, $d^{app}/0.30$, and $g/30$, with a dummy unmatched cost of 1.0. Selected edges also pass the component constraint.

The ablation additionally evaluates geometry-only reciprocal selection and reciprocal linking without the component assertion. A causal diagnostic holds the gated candidate graph and scalar cost fixed while changing only greedy, reciprocal, or partial-Hungarian assignment; this diagnostic is not used to replace the pre-specified method after observing test performance. GIAOTracker is discussed as relevant global-linking work [14], but its authors’ public repository states that code release is not planned and does not provide the learned global-link model and weights needed for a faithful run. We therefore do not label a local approximation as official GIAOTracker.

4.4. Metrics and aggregation

TrackEval commit 12c8791b computes HOTA, its association component AssA, IDF1, CLEAR MOTA, and identity switches (IDSW). HOTA averages the standard localization-threshold curve; AssA measures association quality among matched detections. IDF1 is the F1 score after global one-to-one trajectory matching, and IDSW counts changes in the matched predicted identity over time. Higher is better for HOTA, AssA, IDF1, and MOTA; lower is better for IDSW.

Headline HOTA, AssA, and IDF1 values are equal-sequence means, so long sequences do not dominate. IDSW, FP, and FN are summed over exactly the same sequence set. We never recompute F1 from averaged precision and recall. Paired differences against no refinement and the principal same-input linkers use 10,000 released-sequence bootstrap resamples with seed 20260914. The official release metadata available to us does not map sequence identifiers to source flights, so we do not infer groups from filename prefixes or report an unsupported flight-level block bootstrap. Repeated deterministic executions are reproducibility checks, not independent statistical samples.

For the link audit, an endpoint tracklet receives a post-hoc actor label when at least one observation has a one-to-one same-class GT match at $\text{IoU} \geq 0.9$. The label is the most frequent matched GT identity, with the smaller identity used only as a deterministic tie break. An accepted edge is auditable only when both endpoint labels exist; it is unknown when either endpoint has no qualifying match. Link coverage uses the fixed set of source tracklets with

at least one temporally eligible, non-overlapping successor within 30 frames. Conditional link error is the false-link fraction among auditable accepted edges. Unknown edges remain in coverage but not in this error denominator. A post-hoc appearance-gate grid $\{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ is retained in `risk_coverage.csv`; it is descriptive and is not used to select a test threshold. Table C.13 reports the pre-specified 0.30 operating point.

4.5. Integrity and runtime protocol

Before evaluation, exact GT-as-prediction and globally permuted-ID tests had to return HOTA, IDF1, and MOTA of 100 with IDSW 0. No-refinement had to reproduce the cached tracker output exactly. For every headline refiner, we check preservation of the `(frame, box, confidence, class)` multiset and reject aggregate reporting when one identity appears twice in a frame. Ground-truth actor identity is read only after MMOT predictions are frozen, for metrics and post-hoc edge labels.

Runtime is measured over the full MMOT test set on an NVIDIA RTX 5090 with PyTorch 2.9.1 and CUDA 12.8. BaseReID uses batch 128. Graph timing excludes disk I/O, upstream tracking, descriptor extraction, and metric evaluation; total post-tracking overhead adds descriptor and graph time and divides by 5,466 frames. These are offline replay measurements rather than flight-system latencies.

5. Results

5.1. Primary observation-preserving identity refinement

Table 2 presents the primary verified result under an output contract that preserves observations and permits only identity relabeling. For a given input and upstream tracker, None, AFLink+VA, and CoM3D-ACE receive identical frozen boxes, classes, confidence scores, and local tracklets. CoM3D-ACE has higher equal-sequence-mean IDF1 than both comparators in all six conditions. All twelve paired all-50 intervals in the two gain columns lie above zero.

Relative to no refinement, IDF1 rises by 2.28, 2.29, and 2.18 percentage points for oracle ByteTrack, OC-SORT, and Deep OC-SORT, while total IDSW falls by 2,545, 1,678, and 585. With the shared official detector cache, the corresponding gains are 1.65, 1.62, and 1.00 points and 1,389, 1,059, and 324 fewer switches. Against the box-preserving AFLink adapter, the six IDF1 gains range from 0.75 to 2.02 points. All displayed deltas and intervals are computed from full-precision sequence results before one-step rounding.

Table 2 summarizes gains over unchanged tracker outputs and AFLink+VA. Table 4 extends the comparison to alternative temporal tracklet linkers under the same inputs; Geometry+ReID greedy and partial Hungarian achieve higher mean IDF1 in these conditions.

Table 2: Primary observation-preserving MMOT identity-refinement result over all 50 official test sequences. The upper and lower panels compare CoM3D-ACE with no refinement and AFLink+VA, respectively. IDF1 and its deltas are equal-sequence means. Δ IDSW is CoM3D-ACE minus None over the full release, so negative values indicate fewer switches. Intervals are 95% paired released-sequence bootstraps (10,000 resamples; seed 20260914). All methods receive identical frozen tracklets; AFLink+VA uses the no-GT box-preserving adapter in Section 4. Bold marks CoM3D-ACE and its verified gains.

Input	Tracker	Equal-sequence-mean IDF1		CoM3D-ACE gain [95% CI] vs. None	Δ IDSW vs. None
		None	CoM3D-ACE		
Oracle AABB	ByteTrack	47.44	49.72	+2.28 [+1.78, +2.83]	-2,545
	OC-SORT	42.96	45.25	+2.29 [+1.71, +2.92]	-1,678
	Deep OC-SORT	81.49	83.67	+2.18 [+1.32, +3.29]	-585
Detector	ByteTrack	38.32	39.97	+1.65 [+1.20, +2.13]	-1,389
	OC-SORT	36.46	38.09	+1.62 [+1.21, +2.07]	-1,059
	Deep OC-SORT	63.09	64.08	+1.00 [+0.49, +1.70]	-324

Input	Tracker	Equal-sequence-mean IDF1		CoM3D-ACE gain [95% CI] vs. AFLink+VA
		AFLink+VA	CoM3D-ACE	
Oracle AABB	ByteTrack	47.70	49.72	+2.02 [+1.58, +2.51]
	OC-SORT	43.33	45.25	+1.92 [+1.44, +2.47]
	Deep OC-SORT	81.93	83.67	+1.74 [+0.96, +2.73]
Detector	ByteTrack	38.53	39.97	+1.43 [+1.05, +1.86]
	OC-SORT	36.69	38.09	+1.40 [+1.05, +1.77]
	Deep OC-SORT	63.33	64.08	+0.75 [+0.38, +1.22]

Figure 3 resolves the detector-input means into their 50 sequence-level changes. CoM3D-ACE improves IDF1 on 41, 45, and 32 sequences for ByteTrack, OC-SORT, and Deep OC-SORT, ties on 5, 3, and 9, and worsens on 4, 2, and 9, respectively. The corresponding equal-sequence means are +1.65, +1.62, and +1.00 points. The benefit over unchanged tracker output is therefore distributed across most sequences, although it is not universal and is least consistent for Deep OC-SORT.

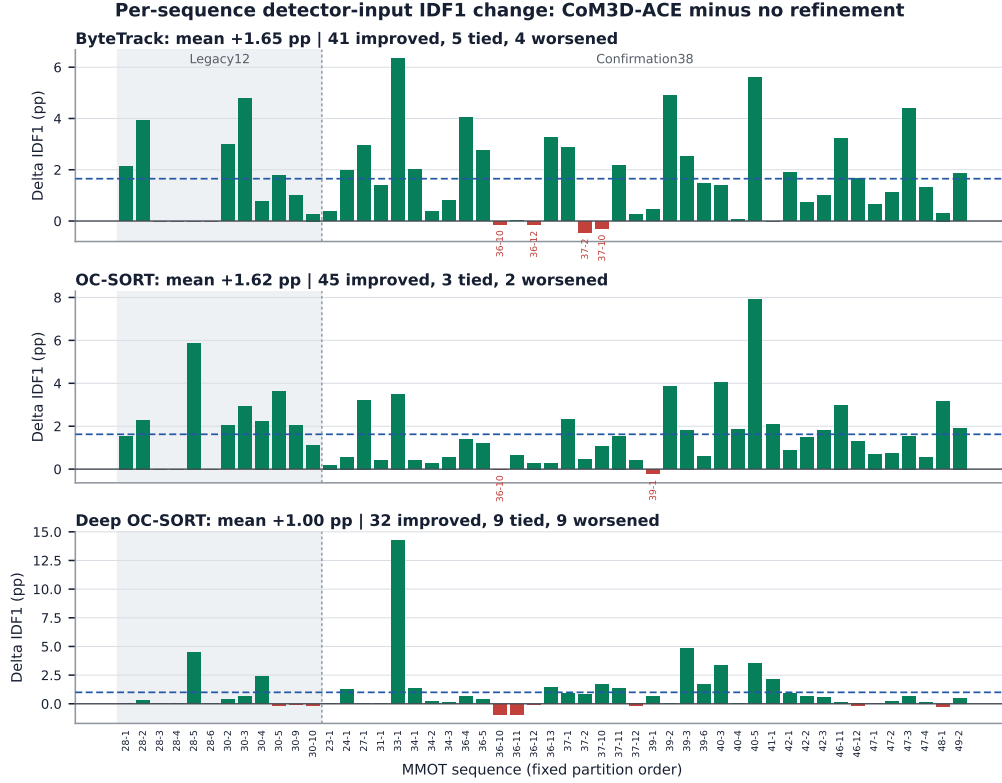


Figure 3: Per-sequence detector-input IDF1 change over all 50 MMOT test sequences. Each bar is CoM3D-ACE minus no refinement in percentage points, and the dashed line is the equal-sequence mean. Panels use the same fixed sequence order; gray shading marks legacy12. Positive, zero, and negative outcomes are all retained. Values come from the same frozen official-detector inputs as Table 2.

The source-stage AFLink audit explains the adapter rather than hiding invalid runs. Frozen StrongSORT source calls native `deduplicate` after identity remapping. Across the six full-test conditions, selected merges create 13 sequence/tracker/class overlap contexts, and native deduplication removes 15 boxes. Decimal save formatting changes numeric serialization separately. The common validity adapter rejects only those overlapping component merges, without GT, and preserves every input box; its valid full-set results appear in Table 2.

The positive findings that remain directly supported are:

- **Upstream preservation:** CoM3D-ACE changes only identity labels; every frame, box, class, confidence, and prediction count is unchanged.

- **No-refinement and AFLink comparisons:** IDF1 improves across three tracker families and both input conditions, with all twelve full-test paired intervals above zero.
- **Protocol breadth:** the direction against no refinement holds with common oracle boxes and one shared official detector output.
- **Training scope:** the refiner has no trainable weights and BaseReID is not fine-tuned on MMOT; the fixed gates were selected on M3OT development streams. This statement does not include the official MMOT-trained detector checkpoint.

5.2. Locally pre-specified reporting partition

The 38 sequences outside the earlier 12-sequence analysis form a locally pre-specified reporting partition. Table 3 reports CoM3D-ACE minus no-refinement changes. All six paired IDF1 intervals remain above zero. With oracle AABBs, 38 and 37 of 38 ByteTrack and OC-SORT sequences improve; with detector inputs, 33, 36, and 27 sequences improve for ByteTrack, OC-SORT, and Deep OC-SORT. These results show that the mean no-refinement gain is not confined to legacy12. Because an externally timestamped preregistration is unavailable, we do not describe this partition as an independent preregistered confirmation set.

Table 3: CoM3D-ACE minus no-refinement paired changes on the locally pre-specified confirmation38 partition. Confidence intervals are 95% released-sequence bootstrap intervals with 10,000 resamples and seed 20260914. The final column gives improved/tied/worsened IDF1 sequence counts.

Input	Tracker	Δ HOTA	Δ AssA	Δ IDF1 (95% CI)	Δ IDSW	I/T/W
Oracle	ByteTrack	+1.05	+1.71	+2.38 [1.85, 2.99]	-2,147	38/0/0
	OC-SORT	+0.91	+2.07	+2.15 [1.54, 2.84]	-1,185	37/0/1
	Deep OC-SORT	+1.28	+2.49	+2.29 [1.37, 3.57]	-466	31/4/3
Official det.	ByteTrack	+0.79	+1.48	+1.71 [1.20, 2.26]	-1,142	33/1/4
	OC-SORT	+0.58	+1.46	+1.51 [1.07, 2.04]	-759	36/0/2
	Deep OC-SORT	+0.44	+0.92	+1.10 [0.48, 2.00]	-268	27/5/6

5.3. Alternative-linker, ranking, and transfer diagnostics

The primary result does not establish universal temporal-linker superiority. Table 4 compares the fixed CoM3D-ACE row with three alternative temporal

tracklet linkers on the same frozen tracker outputs. Geometry+ReID greedy and partial Hungarian have higher mean IDF1 than CoM3D-ACE in all six conditions. Their paired intervals favor the alternative linker in five conditions; only the official-detector Deep OC-SORT intervals include zero (Table B.11). Reporting this comparison separately keeps the verified improvement over unchanged trajectories and AFLink+VA visible while acknowledging the more accurate same-input linkers.

Table 4: Same-input comparison of temporal tracklet linkers on all 50 MMOT test sequences. Entries are equal-sequence-mean IDF1. Every method receives the same frozen tracker outputs, while candidate and ranking rules follow Section 4. Geo.-G, Geo.+ReID-G, and Partial-H denote geometry greedy, geometry-plus-ReID greedy, and partial Hungarian assignment. Bold marks the highest displayed value; complete HOTA, AssA, IDF1, and IDSW results are in Appendix B.

Input	Tracker	CoM3D-ACE	Geo.-G	Geo.+ReID-G	Partial-H
Oracle AABB	ByteTrack	49.72	51.02	51.80	50.78
	OC-SORT	45.25	45.41	46.25	46.19
	Deep OC-SORT	83.67	83.69	83.99	84.00
Detector	ByteTrack	39.97	40.76	41.10	40.56
	OC-SORT	38.09	38.14	38.76	38.74
	Deep OC-SORT	64.08	64.01	64.15	64.17

Figure 4 shows how these aggregate results arise from frozen predictions. In the correct case, CoM3D-ACE joins local IDs 3 and 28, which both receive post-hoc actor label 2, while leaving the intervening ID 1 separate; this is a correct accepted edge, not complete recovery of every fragment. In the failure case, CoM3D-ACE assigns ID 123 to different-actor predecessor 99, whereas partial Hungarian selects same-actor predecessor 49. This example exposes the ranking failure behind the stronger alternative-linker result rather than presenting only favorable cases.



Figure 4: Verified detector-input MMOT tracklet-link examples. Solid blue boxes and local IDs are frozen tracker outputs; dashed orange boxes and identities are post-hoc one-to-one same-class GT audits at $\text{IoU} \geq 0.9$. Identity rows show the final component roots while all boxes remain unchanged. (a) CoM3D-ACE accepts the correct $3 \rightarrow 28$ edge for actor 2; partial Hungarian selects the alternative correct $1 \rightarrow 28$ edge. (b) CoM3D-ACE accepts the false $99 \rightarrow 123$ edge, while partial Hungarian selects the correct $49 \rightarrow 123$ predecessor. Displayed frames are verified observations within the named tracklets; GT is used only after linking for the audit.

The CoM3D-ACE row combines reciprocal selection with the geometry-first lexicographic key in Eq. (3). To separate ranking from solver effects, we reuse each frozen candidate graph and the partial-Hungarian scalar cost with greedy, reciprocal, and Hungarian assignment. At the unchanged 0.30 appearance gate, common-cost reciprocal selection raises IDF1 over CoM3D-ACE in all six conditions. It also accepts a larger fraction of the fixed eligible-source denominator while reducing conditional link error in all six (Tables 5 and C.13). Thus geometry-first ranking is a material limitation; the result does not justify renaming the post-hoc common-cost row as the pre-specified method.

Table 5: Cache-controlled ranking/solver diagnostic on all 50 MMOT sequences. Each cell is equal-sequence-mean IDF1/total IDSW. Geo-first R is the CoM3D-ACE geometry-first reciprocal rule; Cost-G, Cost-R, and Cost-H apply a common scalar cost with greedy, reciprocal, and partial-Hungarian assignment. The fixed 0.30 appearance gate is retained; no test threshold is selected.

Input	Tracker	Geo-first R	Cost-G	Cost-R	Cost-H
Oracle	ByteTrack	49.72/17,511	51.54/14,929	50.30/16,710	50.78/16,462
	OC-SORT	45.25/4,441	46.14/3,501	45.88/3,828	46.19/3,547
	Deep OC-SORT	83.67/1,377	83.96/1,185	84.02/1,242	84.00/1,174
Detector	ByteTrack	39.97/8,935	40.97/7,604	40.42/8,491	40.56/8,405
	OC-SORT	38.09/3,204	38.66/2,635	38.55/2,825	38.74/2,668
	Deep OC-SORT	64.08/1,046	64.17/945	64.23/979	64.17/957

The reciprocal path invariant was also checked directly. Across 1,989 MMOT sequence/class/tracker/input instances, maximum proposal indegree and outdegree are one, with no temporal violation, cycle, duplicate-frame input, component rejection, or guard/no-guard output difference. This confirms that the component check is defensive under the implemented one-shot graph rather than a measured source of accuracy gain.

M3OT provides an adverse transfer diagnostic. The fixed-rule rows reproduce exactly, including accepted-edge sets. On held-out streams, CoM3D-ACE accepts 0 correct and 7 false links for ByteTrack and 0 correct and 3 false links for OC-SORT. Among the eight auditable same-actor opportunities, five fail the 55-pixel gate, two lose the reciprocal incoming rank, and one loses the outgoing rank. Common-cost reciprocal recovers one correct link per tracker and reduces false links to 6 and 2, but IDF1 remains below no refinement (Table D.15). The failure therefore involves both ranking and non-transferring

cues, not only the choice of reciprocal versus Hungarian assignment.

5.4. Small-object context and execution cost

Table 6 verifies that the evaluation contains many small aerial targets. Across all 50 sequences, 53.4% of boxes have area below 32^2 pixels, 45.9% lie from 32^2 to 96^2 , and the median short side is 21 pixels. This distribution motivates the setting but is not a scale-stratified claim that CoM3D-ACE improves tiny objects more than larger objects.

Table 6: Native MMOT annotation-box distribution. Area bins are fixed COCO boundaries; Side denotes the shorter AABB side, and percentages use all valid boxes in each partition.

Partition	Boxes	$A < 32^2$	$32^2 \leq A < 96^2$	Side < 16	Median side
All 50	196,194	53.4%	45.9%	43.1%	21 px
Legacy 12	30,666	47.9%	51.7%	38.5%	23 px
Confirm. 38	165,528	54.4%	44.9%	44.0%	19 px

Post-tracking runtime is dominated by appearance extraction (Table 7). BaseReID plus graph processing costs 46.9 ms/frame for ByteTrack and approximately 24.5 ms/frame for OC-SORT and Deep OC-SORT. Graph construction and decisions alone total 7.99, 2.09, and 1.21 seconds, respectively, over the 5,466-frame release. A separate same-machine CPU replay shows lower reciprocal solver time and peak memory than Hungarian in most conditions, but the absolute solver difference is small relative to candidate construction and descriptor extraction (Table C.14). It is not an end-to-end runtime advantage.

Table 7: Full-test post-tracking runtime. Crops counts sampled tracker-box crops (up to first, middle, and last per tracklet), Desc. counts aggregated tracklet descriptors, and Cand. counts pre-gate temporal candidate pairs. ReID uses an RTX 5090 and batch 128; graph time is in-memory CPU candidate construction and decision time. Total combines ReID and graph time over 5,466 frames.

Tracker	Crops	Desc.	ReID (s)	Cand.	Graph (s)	Total (ms/fr.)
ByteTrack	29,281	13,836	248.3	864,743	7.99	46.9
OC-SORT	15,819	8,150	131.6	216,508	2.09	24.5
Deep OC-SORT	15,490	5,663	132.9	76,253	1.21	24.5

6. Discussion

The clearest positive result is a protocol-matched improvement over unchanged upstream trajectories. CoM3D-ACE raises the equal-sequence means of HOTA, AssA, and IDF1 and lowers total IDSW for three tracker families under both oracle and official-detector inputs. Because the method preserves the complete detection multiset, these changes are attributable to identity relabeling rather than detector retraining, interpolation, or sequence filtering. The locally pre-specified confirmation³⁸ analysis shows that this mean gain is not confined to the 12 sequences used in the earlier analysis. CoM3D-ACE also improves IDF1 over the validated box-preserving AFLink adapter in all six full-test conditions; all 12 full-test paired intervals against no refinement and AFLink+VA lie above zero.

That evidence establishes improvement over unchanged trajectories and the validated AFLink adapter, not superiority among temporal linkers. The same-input comparison and paired intervals show that Geometry+ReID greedy and partial Hungarian produce higher mean IDF1 than the fixed CoM3D-ACE ranking in all six conditions. Moreover, replacing only geometry-first ranking with the common scalar cost improves reciprocal assignment itself. The evidence therefore supports reciprocal selection as an auditable deterministic mechanism, but not the fixed rank key as an accuracy- or risk-optimal design.

The EvidenceToken abstraction remains operationally useful because it decouples refinement from a particular tracker implementation. ByteTrack, OC-SORT, and Deep OC-SORT expose different motion and appearance behavior, yet the same candidate graph can process their archived tracklets. Unsupported fragments remain unchanged, every merge is recoverable from an edge trace, and the native AFLink failure illustrates why preserving an explicit output contract matters.

The transfer and runtime audits further delimit this role. M3OT shows that a fixed pixel gate and MOT17 appearance descriptor do not transfer reliably, even when a different solver recovers some correct candidates. Appearance extraction and candidate construction also dominate assignment time, so a modest reciprocal-versus-Hungarian solver saving is not a practical end-to-end advantage. These findings motivate future development-set redesign while keeping the present test results unchanged.

7. Limitations

The verified claim is deliberately bounded to deterministic temporal refinement of frozen trajectories within each UAV stream. The present protocol does not evaluate persistent identity across cameras, 3D reconstruction, coordinated view planning, or active re-observation; those tasks require separate identity and action-level ground truth.

The method is not a universally optimal temporal linker. Geometry+ReID greedy and partial Hungarian have higher full-test mean IDF1 than fixed CoM3D-ACE in all six same-input conditions. Common-cost reciprocal assignment also improves both IDF1 and the observed error-coverage pair over the fixed geometry-first ranking in all six conditions. Accordingly, the current contribution is an auditable fixed refiner and controlled diagnosis, not a state-of-the-art linker claim. Any revised ranking must be designed on development data and evaluated on new confirmation data rather than selected from these test results.

The component-overlap check is a defensive assertion rather than an independent performance module. Strict forward-time reciprocal proposals form disjoint paths when input tracklets contain no duplicate frame, so the check cannot reject a proposal under the evaluated one-shot rules. Accepted-link error is defined only where both endpoint identities are auditable. At the fixed operating point, the auditable share is 4.4–13.8% for detector-input links, 12.8–13.3% for oracle ByteTrack, and 100% for oracle OC-SORT and Deep OC-SORT. Unknown links remain in coverage, and these conditional rates are not calibrated risk estimates.

Cross-dataset transfer remains unresolved. The unchanged MMOT-beneficial rule improves four M3OT development streams but degrades four held-out streams. Five of eight held-out same-actor opportunities fail the fixed pixel gate, and common-cost reciprocal selection recovers only one correct link per tracker while remaining below no refinement. The M3OT descriptor cache also uses an oracle-GT IoU crop-admission rule, which limits it to a diagnostic rather than a detector-input transfer claim.

Native AFLink deduplication resolves overlapping remapped identities by deleting boxes and therefore does not satisfy the common-box contract. We report a separately named no-GT validity adapter instead of calling it unmodified official output. A faithful GIAOTracker run remains blocked by unavailable released implementation and compatible weights. Finally, the MMOT release metadata available to us does not expose source-flight

groupings, the local confirmation split lacks external preregistration evidence, and runtime is an offline replay on one workstation rather than an onboard latency or energy claim.

8. Conclusion

CoM3D-ACE provides a deterministic evidence-graph refiner for frozen aerial tracklets. Relative to no refinement, it improves equal-sequence-mean HOTA, AssA, and IDF1 and reduces total identity switches over 50 MMOT sequences for ByteTrack, OC-SORT, and Deep OC-SORT under both oracle and official-detector inputs. IDF1 also exceeds the validated box-preserving AFLink adapter in all six conditions, and all 12 full-release paired intervals against no refinement and AFLink+VA lie above zero. The no-refinement IDF1 difference remains positive on the locally pre-specified 38-sequence partition, and every input box, class, confidence, and prediction count is preserved.

The same-input temporal-linker comparison establishes the boundary of that result: the fixed CoM3D-ACE ranking is not accuracy-optimal, a common-cost test identifies geometry-first ranking as a limitation, and held-out M3OT transfer is negative. CoM3D-ACE should therefore be read as a reproducible, auditable MMOT refinement and diagnostic framework, not as a universally superior association rule. Cross-camera identity, active observation, and redesigned cross-domain ranking remain future work requiring new, independently evaluated protocols.

CRedit authorship contribution statement

Jong-Chan Park: Conceptualization, Methodology, Investigation, Writing – original draft. So-Hee Lim: Visualization, Writing – review & editing. Sujin Park: Visualization, Writing – review & editing. Sang-Min Choi: Formal analysis, Writing – review & editing. Gun-Woo Kim: Formal analysis, Writing – review & editing.

Data and code availability

The MMOT [10] and M3OT [11] datasets are publicly available from their official sources: the MMOT repository, the MMOT dataset page, and the M3OT repository. The implementation, frozen split manifest, configurations,

and reproduction instructions are planned for public release through a GitHub repository. No repository URL is claimed before that release is available.

Appendix A. MMOT reporting partitions

Table A.8 records the locally pre-specified reporting split. The confirmation38 partition contains every official test sequence not in the previously analyzed legacy12 set.

Table A.8: Locally pre-specified MMOT reporting partition. Frame counts and source hashes are stored in the accompanying split manifest. No sequence was assigned by performance; no external preregistration is claimed.

Partition	Sequence identifiers
Legacy12	data28-1, data28-2, data28-3, data28-4, data28-5, data28-6, data30-2, data30-3, data30-4, data30-5, data30-9, data30-10
Confirm. 38	data23-1, data24-1, data27-1, data31-1, data33-1, data34-1, data34-2, data34-3, data36-4, data36-5, data36-10, data36-11, data36-12, data36-13, data37-1, data37-2, data37-10, data37-11, data37-12, data39-1, data39-2, data39-3, data39-6, data40-3, data40-4, data40-5, data41-1, data42-1, data42-2, data42-3, data46-11, data46-12, data47-1, data47-2, data47-3, data47-4, data48-1, data49-2

Appendix B. Full-metric and paired linker comparisons

Tables B.9 and B.10 provide complete HOTA, AssA, IDF1, and IDSW values for the operational comparison in Table 2 and the alternative-linker comparison in Table 4. All rows receive the same saved tracklets within each upstream/input condition.

Table B.11 reports paired all-50 differences rather than relying only on aggregate means.

Table B.9: MMOT full-test temporal refinement with common oracle AABBs. HOTA, AssA, and IDF1 are equal-sequence means over 50 sequences; IDSW is the total. Bold values mark the best valid refiner for each upstream tracker. AFLink+VA uses the fixed no-GT output-validity adapter described in Section 4; Geo. and Hung. abbreviate geometry and Hungarian.

Upstream	Refiner	HOTA	AssA	IDF1	IDSW
ByteTrack	None	44.20	44.69	47.44	20,056
	Geometry greedy	45.60	46.97	51.02	15,484
	Geo.+ReID greedy	45.89	47.47	51.80	15,064
	Partial Hung.	45.55	46.90	50.78	16,462
	AFLink+VA	44.30	44.88	47.70	20,025
	CoM3D-ACE	45.19	46.34	49.72	17,511
OC-SORT	None	44.92	51.43	42.96	6,119
	Geometry greedy	45.90	53.68	45.41	4,122
	Geo.+ReID greedy	46.31	54.55	46.25	3,481
	Partial Hung.	46.26	54.45	46.19	3,547
	AFLink+VA	45.14	51.92	43.33	6,082
	CoM3D-ACE	45.92	53.64	45.25	4,441
Deep OC-SORT	None	81.32	82.06	81.49	1,962
	Geometry greedy	82.58	84.52	83.69	1,294
	Geo.+ReID greedy	82.82	84.98	83.99	1,149
	Partial Hung.	82.79	84.93	84.00	1,174
	AFLink+VA	81.54	82.48	81.93	1,919
	CoM3D-ACE	82.60	84.54	83.67	1,377

Table B.10: MMOT full-test temporal refinement from one shared official YOLO11L-3ch detection cache. Aggregation and invariant handling match Table B.9.

Upstream	Refiner	HOTA	AssA	IDF1	IDSW
ByteTrack	None	34.65	42.29	38.32	10,324
	Geometry greedy	35.64	44.14	40.76	7,908
	Geo.+ReID greedy	35.79	44.49	41.10	7,711
	Partial Hung.	35.58	44.11	40.56	8,405
	AFLink+VA	34.73	42.45	38.53	10,297
	CoM3D-ACE	35.40	43.75	39.97	8,935
OC-SORT	None	35.89	47.62	36.46	4,263
	Geometry greedy	36.51	49.40	38.14	2,959
	Geo.+ReID greedy	36.80	50.12	38.76	2,585
	Partial Hung.	36.77	49.97	38.74	2,668
	AFLink+VA	36.02	47.95	36.69	4,240
	CoM3D-ACE	36.53	49.24	38.09	3,204
Deep OC-SORT	None	53.92	67.00	63.09	1,370
	Geometry greedy	54.25	67.71	64.01	1,008
	Geo.+ReID greedy	54.34	67.92	64.15	942
	Partial Hung.	54.36	67.95	64.17	957
	AFLink+VA	54.00	67.16	63.33	1,342
	CoM3D-ACE	54.31	67.84	64.08	1,046

Table B.11: CoM3D-ACE minus same-input comparators in IDF1, with 95% paired released-sequence bootstrap intervals (10,000 resamples; seed 20260914). The upper panel compares alternative temporal linkers: the two official-detector Deep OC-SORT intervals include zero and the other ten are below zero. The lower panel compares AFLink+VA: all six intervals are above zero, and W/T/L counts sequence-level IDF1 wins, ties, and losses for CoM3D-ACE.

Input	Tracker	vs. Geo.+ReID greedy	vs. partial Hung.
Oracle	ByteTrack	-2.08 [-2.69, -1.53]	-1.07 [-1.37, -0.79]
	OC-SORT	-1.00 [-1.38, -0.66]	-0.93 [-1.36, -0.57]
	Deep OC-SORT	-0.32 [-0.50, -0.15]	-0.33 [-0.51, -0.17]
Detector	ByteTrack	-1.14 [-1.85, -0.60]	-0.60 [-0.82, -0.40]
	OC-SORT	-0.68 [-1.01, -0.41]	-0.66 [-0.99, -0.40]
	Deep OC-SORT	-0.06 [-0.14, +0.01]	-0.09 [-0.19, +0.01]

Input	Tracker	vs. AFLink+VA	IDF1 W/T/L
Oracle	ByteTrack	+2.02 [+1.58, +2.51]	46/4/0
	OC-SORT	+1.92 [+1.44, +2.47]	45/3/2
	Deep OC-SORT	+1.74 [+0.96, +2.73]	39/8/3
Detector	ByteTrack	+1.43 [+1.05, +1.86]	43/5/2
	OC-SORT	+1.40 [+1.05, +1.77]	44/3/3
	Deep OC-SORT	+0.75 [+0.38, +1.22]	32/9/9

Appendix C. Component and accepted-link diagnostics

Table C.12 separates the fixed graph inputs. Geometry-only reciprocal selection accounts for most of the no-refinement gain, and the appearance gate adds a smaller increment. The no-guard and full rows are identical for the structural path reason established in Section 3, not because an independently active module happens to have no effect.

Table C.12: Temporal component ablation on all 50 MMOT sequences with oracle AABBs. IDF1 is the equal-sequence mean and IDSW is the total. All rows reuse the same upstream tracklets and candidate universe.

Variant	ByteTrack		OC-SORT		Deep OC-SORT	
	IDF1	IDSW	IDF1	IDSW	IDF1	IDSW
None	47.44	20,056	42.96	6,119	81.49	1,962
Geometry reciprocal	49.62	17,581	45.08	4,551	83.63	1,392
+ReID reciprocal, no guard	49.72	17,511	45.25	4,441	83.67	1,377
CoM3D-ACE	49.72	17,511	45.25	4,441	83.67	1,377

Table C.13 reports the fixed operating point for the geometry-first and common-cost reciprocal rankings. Coverage uses the fixed eligible-source denominator. Auditable is the fraction of accepted links for which both endpoint tracklets receive post-hoc majority-GT labels. An endpoint is unlabeled when none of its observations has a one-to-one same-class GT match at $\text{IoU} \geq 0.9$; any accepted edge containing such an endpoint is unknown. Conditional error is therefore evaluated only over auditable accepted links. The rates are descriptive and do not define calibrated risk.

Table C.13: Full-test link audit at the unchanged 0.30 appearance gate. Coverage is accepted/eligible-source; auditable is (correct+false)/accepted; conditional error is false/(correct+false). C/F/U gives correct, false, and unknown accepted links. Unknown links remain in coverage but not the error denominator. Cost-R is a post-hoc causal diagnostic, not a selected replacement method.

Input	Tracker	Rule	Coverage	Auditable	Cond. error	C/F/U
Oracle	ByteTrack	Geo-first R	36.6%	12.8%	44.6%	317/255/3,909
		Cost-R	39.8%	13.3%	43.2%	369/281/4,226
	OC-SORT	Geo-first R	37.9%	100.0%	35.1%	1,661/899/0
		Cost-R	42.1%	100.0%	21.2%	2,239/602/0
	Deep OC-SORT	Geo-first R	26.1%	100.0%	39.2%	575/371/0
		Cost-R	27.4%	100.0%	29.0%	704/287/0
Detector	ByteTrack	Geo-first R	34.4%	4.7%	29.7%	97/41/2,796
		Cost-R	37.2%	4.4%	24.8%	106/35/3,033
	OC-SORT	Geo-first R	35.4%	13.8%	38.9%	173/110/1,766
		Cost-R	39.0%	13.3%	31.6%	206/95/1,954
	Deep OC-SORT	Geo-first R	21.7%	8.4%	32.3%	42/20/675
		Cost-R	22.1%	8.8%	28.8%	47/19/686

Table C.14: Cache-only solver replay on an Intel Xeon Silver 4510 CPU. Times sum instance means over the full release; peak memory is the maximum Python allocation traced for one solver call. Geo-first R and Cost-H reuse the same candidate graph. Cand. is candidate-graph construction. Descriptor extraction and AFLink model inference are excluded.

Input	Tracker	Cand. (ms)	Geo-first R (ms/MB)	Cost-H (ms/MB)
Oracle	ByteTrack	8,263.7	191.5/3.9	325.8/17.9
	OC-SORT	1,965.0	82.9/3.6	102.1/3.6
	Deep OC-SORT	1,086.7	118.1/6.6	127.9/6.7
Detector	ByteTrack	3,153.8	133.7/4.2	184.4/7.1
	OC-SORT	1,408.3	77.8/3.8	92.8/3.8
	Deep OC-SORT	817.1	95.7/4.3	101.7/4.4

Appendix D. Cross-dataset transfer diagnostic

The same temporal, image-plane, appearance, and reciprocal settings were fixed on four M3OT development streams and then applied unchanged to four held-out streams. The fixed outputs and edge sets reproduce exactly. Of eight held-out same-actor opportunities before the geometry/appearance gates, five fail the geometry gate, two lose incoming rank, and one loses outgoing rank. Table D.15 compares unchanged inputs under fixed alternative solvers.

Table D.15: Fixed-policy M3OT development and held-out diagnostic. The upper table reports equal-sequence-mean IDF1/total IDSW; the lower reports post-hoc accepted correct/false links. Geo-first R is CoM3D-ACE reciprocal ranking; Cost-G, Cost-R, and Cost-H share the same scalar cost. Tracker inputs are oracle AABBs. Descriptor observations are admitted by one-to-one oracle-GT IoU ≥ 0.9 matching and then cropped at the tracker boxes.

Split	Tracker	None	Geo-first R	Cost-G	Cost-R	Cost-H
Development	ByteTrack	96.89/20	98.49/10	98.47/11	98.47/11	98.55/9
	OC-SORT	96.10/21	98.26/9	98.24/10	98.24/10	98.33/6
Held-out	ByteTrack	96.77 /6	92.77/6	92.89/5	92.89/5	91.86/6
	OC-SORT	95.39 /5	94.84/5	94.92/4	94.92/4	93.96/5

Split	Tracker	Geo-first R	Cost-G	Cost-R	Cost-H
Development	ByteTrack	10/3	9/5	9/4	11/4
	OC-SORT	12/5	11/7	11/6	15/4
Held-out	ByteTrack	0/7	1/6	1/6	0/8
	OC-SORT	0/3	1/2	1/2	0/4

Appendix E. Reproducibility record

The release manifest records full revisions and checksums. Their compact prefixes are:

- MMOT test revision: 57dfe1521ee2;
- BaseReID SHA-256: 4bf6b63d4923;
- MMOT YOLO11L-3ch SHA-256: 6d9f66ed17eb;
- AFLink SHA-256: b35cbecdd3ac;
- TrackEval commit: 12c8791b303e;
- BoxMOT commit: 6edfa8ad7dc2.

Evaluator sanity tests return HOTA/IDF1/MOTA 100 and IDSW 0 for both exact GT predictions and a global one-to-one permutation of prediction identities. Legacy12 rows in the standalone and full-50 executions match at absolute tolerance zero. The source artifact package contains commands, source and model hashes, successful manifests, failed-run records, per-sequence CSVs, and a SHA-256 index.

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